

Complex Arithmetic

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$\mathbb{R}$: set of real numbers

$\mathbb{C}$: set of complex numbers; i.e. $(a + ib)$ if a, b are real, and i is imaginary.

 i and $-i$ are roots of $x^2 + 1 = 0$;
i.e. $i^2 = (-i)^2 = -1$.

The powers of i are cyclical:

$$i^4 = (-1)(-1) = 1 ,$$

$$i^5 = i^{4+1} = 1 * i = i , \text{ and}$$

$$(i^{-1})(i) = 1 \text{ (where } i^{-1} \text{ is the multiplicative inverse of } i).$$

Complex number: $a + ib$

Complex conjugate: $a - ib$

- Complex roots to polynomials that have real coefficients always have complex conjugate pairs.
- Complex conjugates are denoted by *: $(a + ib)^*$ or bar: $\overline{(a + ib)}$.

Addition and subtraction:

$$(a + ib) + (c + id) = (a + c) + i(b + d).$$

The complex conjugate of a sum is the sum of the complex conjugates.

Multiplication:

$$(a + ib)(c + id) = (ac - bd) + i(ad + bc).$$

If α is a complex number,

$$\alpha\alpha^* =: |\alpha|^2 .$$

$$\text{If } \alpha = a + ib , |\alpha|^2 = a^2 + b^2 .$$

Modulus (absolute value of α):

$|\alpha| = +\sqrt{a^2 + b^2}$ (always real & non-negative).

Inverse of a complex number: $\alpha^{-1} = \alpha^* / |\alpha|^2$.

The inverse/negative-first power of i (a.k.a. $(-i)^2$) is $-i$.

- This means that $\alpha * \alpha^{-1} = \alpha^{-1} * \alpha = 1$.

Division (multiplication by an inverse):

$$\alpha \div \beta = \frac{\alpha}{\beta} = \beta^{-1} \alpha.$$

To rationalize an irrational fraction (a fraction that has a complex denominator), multiply both the numerator and denominator by the denominator's complex conjugate.

When we graph complex numbers, they are put in the "Complex" or "Argand" Plane, where

x : $\mathbb{R}$, and

y : $\mathbb{I}$ ($\text{Im}(z)$).

Can be represented in polar coordinates (θ, r) .

For $\alpha = a + ib$,

$$a = r \cos(\theta) ,$$

$$b = r \sin(\theta) .$$

$\alpha = a + ib =: r e^{i\theta} = r(\cos(\theta) + i \sin(\theta))$, where

$r = \sqrt{a^2 + b^2}$ (the modulus of α), and

$\theta = \arctan(\frac{b}{a})$ (the argument/phase of α) .

$$\alpha_1 \alpha_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)} ,$$

$$\frac{\alpha_1}{\alpha_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)} .$$

★ Multiple-angle formulae:

$$\cos(\theta_1 + \theta_2) = \cos(\theta_1) \cos(\theta_2) - \sin(\theta_1) \sin(\theta_2) ,$$

$$\sin(\theta_1 + \theta_2) = \sin(\theta_1) \cos(\theta_2) + \cos(\theta_1) \sin(\theta_2) .$$

Considering $e^{in\theta}$,

★ $(\cos(\theta) + i \sin(\theta))^n = \cos(n\theta) + i \sin(n\theta)$ *DeMoivre's Formula.

If we write $\alpha = r e^{i\theta}$, we can find $\alpha^{p/q}$ (where p, q are integers) by $\alpha^{p/q} = r^{p/q} e^{ip\theta/q}$; but since

$$e^{i(\theta+2n\pi)} = e^{i\theta} e^{2ni\pi} = e^{i\theta} 1 ,$$

the rational root is multiple-valued with q different roots:

$$\alpha^{p/q} = r^{p/q} e^{ip(\theta+2n\pi)/q}$$

$$= r^{p/q} (\cos(p(\theta+2n\pi)/q) + i \sin(p(\theta+2n\pi)/q)) \text{ for } n = 0, 1, 2, \dots, q-1 .$$

An irrational power of a complex number has infinite values.

$$\alpha = re^{i\theta} \rightarrow \text{since } \cos(\theta+2n\pi) = \cos(\theta) \text{ and}$$

$$\sin(\theta+2n\pi) = \sin(\theta) , \text{ then}$$

$$e^{i(\theta+2n\pi)} = e^{i\theta} , \text{ and}$$

$$\ln \alpha = \ln r + i(\theta+2n\pi).$$

- Infinitely many-valued; that for which $n = 0$ is called the "principal value".

Natural log rules:

$$e^{\ln \alpha} = \alpha ,$$

$$\ln(z_1 z_2) = \ln(z_1) + \ln(z_2) ,$$

$$\ln(z_1/z_2) = \ln(z_1) - \ln(z_2) , \text{ and}$$

$$\ln(z^\alpha) = \alpha \ln z .$$

Since $a^\alpha = e^{\ln(a^\alpha)} = e^{\alpha \ln a}$; then

$$e^{a+b} = e^a e^b ,$$

$$e^\alpha = e^{a+ib} = e^a e^{ib} = e^a (\cos(b) + i \sin(b)) .$$

$$\star \cos \alpha = \frac{e^{i\alpha} + e^{-i\alpha}}{2} ,$$

$$\star \sin \alpha = \frac{e^{i\alpha} - e^{-i\alpha}}{2i} .$$

When $a = 0$,

$$\star \cos(ib) = \frac{e^{-b} + e^b}{2} = \frac{e^b + e^{-b}}{2} = \cosh b , \text{ and}$$

$$\star \sin(ib) = \frac{-e^b - e^{-b}}{2i} = i \frac{e^b + e^{-b}}{2} = i \sinh b \text{ (cosh and sinh being hyperbolic cosine and sine), so}$$

$$\cosh(ia) = \cos(a) , \text{ and}$$

$$\sinh(ia) = i \sin(a) .$$